



Spring Boot – Interview Questions

1. What is Spring Boot and how is it different from Spring?

Cross Q1: What problems in Spring does Spring Boot solve?

Cross Q2: Is Spring Boot a replacement or extension of Spring Framework?

2. What is Spring Boot Starter?

Cross Q1: Give an example of commonly used starter dependencies.

Cross Q2: How does Spring Boot know which starter to use?

3. What is auto-configuration in Spring Boot?

Cross Q1: How does Spring Boot know what configuration to apply?

Cross Q2: How can you disable auto-configuration for a specific class?

4. What is the purpose of `@SpringBootApplication`?

Cross Q1: What are the annotations internally combined in `@SpringBootApplication`?

Cross Q2: Can this annotation be used in multiple classes?

5. How do you create a RESTful API in Spring Boot?

Cross Q1: What annotations are required for a REST controller?

Cross Q2: What is the default response format of a REST API in Spring Boot?

6. What is the difference between `@Controller` and `@RestController`?

Cross Q1: Which one automatically applies `@ResponseBody`?

Cross Q2: Can we use `@ResponseBody` with `@Controller`?

7. How do you connect Spring Boot with a database?

Cross Q1: What starter is used for JPA with Spring Boot?

Cross Q2: Where do you configure database properties?

8. What is `application.properties` and `application.yml`?

Cross Q1: Which format is preferred for hierarchical configuration?

Cross Q2: How can you access properties in a Java class?

9. What is Spring Boot DevTools?

Cross Q1: Does DevTools work with jar files in production?

Cross Q2: How does it support live reloading?



10. What is Spring Boot Actuator?

Cross Q1: What endpoints does Actuator expose by default?

Cross Q2: How can you secure or customize Actuator endpoints?

11. What is the role of CommandLineRunner and ApplicationRunner?

Cross Q1: When do these interfaces get executed?

Cross Q2: Which one gives access to raw command-line arguments?

12. How do you implement exception handling in Spring Boot REST API?

Cross Q1: What is the use of @ControllerAdvice?

Cross Q2: How do you return a custom error response?

13. How do you use profiles in Spring Boot?

Cross Q1: How do you activate a profile via properties or command line?

Cross Q2: Can one class belong to multiple profiles?

14. What is the use of @ConfigurationProperties?

Cross Q1: How is it different from @Value?

Cross Q2: How do you bind complex nested properties?

15. What is the use of @EnableAutoConfiguration?

Cross Q1: Is this annotation mandatory when using @SpringBootApplication?

Cross Q2: What is spring.factories in relation to this?

16. How do you schedule tasks in Spring Boot?

Cross Q1: What annotation enables scheduling?

Cross Q2: Can you schedule tasks with cron expressions?

17. What is Spring Boot's embedded server?

Cross Q1: Which servers does Spring Boot support?

Cross Q2: How do you change the default server port?

18. How do you build and run a Spring Boot application as a jar?

Cross Q1: What is the use of spring-boot-maven-plugin?

Cross Q2: Can you run Spring Boot apps without installing Tomcat?

19. What is the purpose of @Entity, @Repository, and @Service in Spring Boot + JPA?

Cross Q1: How does Spring handle data access exceptions with @Repository?

Cross Q2: Which layer is responsible for business logic?



20. What is pagination in Spring Data JPA?

Cross Q1: Which interface helps with pagination?

Cross Q2: How do you return only a page of data in a REST API?

21. How do you create a custom query in Spring Data JPA?

Cross Q1: How do you use @Query annotation?

Cross Q2: Can you write native SQL queries in Spring Data?

22. What is the difference between JpaRepository, CrudRepository, and PagingAndSortingRepository?

Cross Q1: Which one provides more functionality?

Cross Q2: Can we create our own custom base repository?

23. What is the difference between @PathVariable and @RequestParam?

Cross Q1: When would you prefer one over the other?

Cross Q2: Can both be used in the same method?

24. How do you validate user input in Spring Boot?

Cross Q1: What is the role of @Valid and @NotNull?

Cross Q2: Where do you define validation rules?

25. What are the common ways to secure a Spring Boot application?

Cross Q1: What is Spring Security? Can it be integrated easily with Spring Boot?

Cross Q2: How do you protect REST endpoints using role-based security?

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